

Varun Kumar Atkuri

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PROFESSIONAL SUMMARY:

Data Engineer with 5 years of experience building cloud-native data pipelines across healthcare and retail domains. Skilled in Snowflake (Snowpark), dbt, Python, SQL, and AWS (S3, Glue, Kinesis, Step Functions). Hands-on experience in developing modular data models, implementing incremental ELT workflows, and supporting analytics and machine learning teams with high-quality, HIPAA-compliant datasets. Strong focus on performance, cost-efficiency, and data governance best practices. Comfortable collaborating with cross-functional teams in agile environments.

PROFESSIONAL EXPERIENCE:

McKesson – Data Engineer | Atlanta, GA | Aug 2024 – Present

Project: *Enterprise Healthcare Data Lake and Analytics Platform (AWS Cloud)*

- Built layered **dbt** models in **Snowflake**, improving query modularity and reuse across analytics and **ML pipelines**.
- Produced ML-ready training and validation datasets for clinical, claims, and research models, supporting disease prediction, patient stratification, and risk scoring.
- Engineered point-in-time features (as-of joins, label fences, look-back windows) to prevent data leakage in supervised ML pipelines and ensure model generalizability.
- Implemented incremental dbt pipelines for large fact tables with surrogate keys and late-arriving logic, reducing recomputation and improving job runtimes by 18%.
- Developed Snowpark UDF/UDTFs in **Python** to extract **HL7** fields, normalize ICD/CPT codes, and compute rolling metrics, enabling efficient in-warehouse feature engineering.
- Built ingestion pipelines from S3 to **AWS Glue** to Snowflake for CSV/HL7/XML formats; configured **Snowpipe** for real-time priority feeds supporting operational models.
- Added data quality safeguards using **dbt tests** and custom outlier checks; built quarantine flows that decreased feature defects in two tables by 12%.
- Produced versioned feature tables/views with change logs and column contracts; validated export parity to ensure reproducibility of ML feature snapshots.
- Automated ML feature exports from Snowflake to S3 for SageMaker batch and real-time training workflows; partnered with ML teams to iterate on model features.
- Orchestrated jobs in **Apache Airflow** integrating dbt, Snowpipe, Snowflake Tasks, and export steps; added CloudWatch metrics and SLA checks.
- Applied **HIPAA-aligned controls** using IAM/KMS, Snowflake RBAC, and row/masking policies; performed de-identification in Snowpark before data leaves the warehouse.
- Staged ML model scores and inference results in Snowflake marts to support **SageMaker batch inference** and **Tableau operational dashboards**.

Tech Stack: *Python, SQL, dbt, Snowflake (Snowpipe, Streams/Tasks, RBAC, masking, Snowpark), AWS (S3, Glue, IAM, KMS, CloudWatch), Apache Airflow, Tableau, Git/GitHub, Jupyter Notebook*

Nisum Technologies – Data Engineer | Hyderabad, India | Nov 2021 – Aug 2023

Project: Healthcare Batch & Streaming ETL on AWS

- Contributed to architectural design and implementation of scalable batch and streaming pipelines across 15+ healthcare data sources, integrating analytics and ML workloads.
- Ingested SFTP files and MongoDB exports into **Amazon S3** (landing/raw zones), partitioned by source/date for optimized downstream processing.
- Captured real-time clinical and IoT telemetry via **Amazon Kinesis** Streams and **Firehose** → S3, enabling low-latency data ingestion for operational use cases.
- Engineered **PySpark** transformations in AWS Glue to standardize healthcare schemas (encounters, claims, diagnoses), **processing 3M+ records/day**.
- Applied **ETL** logic for type casting, deduplication, schema drift handling, and late-arriving data using metadata-driven Glue configurations.
- Modeled **Lakehouse architecture** (bronze/silver/gold) in Parquet, integrated with the **Glue Data Catalog** for discoverability and cross-service querying.
- Tuned **Firehose** buffering and S3 partitioning, reducing small-file issues by 35%, improving Redshift COPY efficiency by 40%, and lowering Athena query cost by 20%.
- Built Redshift marts with optimized DIST/SORT keys; enabled ad-hoc analytics in **Athena** and **Tableau** for 6+ business teams.
- Orchestrated pipelines with **AWS Step Functions** and **EventBridge**, adding retry logic, CloudWatch metrics, and SLA tracking dashboards.
- Implemented Glue Data Quality and PySpark-based validation (null, uniqueness, referential integrity), reducing DQ defects by 20%.
- Integrated **CI/CD pipelines** using **GitHub Actions** for automated Glue job deployment, **Terraform** for IaC provisioning (S3, IAM, Redshift), and **CloudFormation** templates for ETL orchestration resources.
- Containerized ETL microservices with **Docker** and deployed Glue development utilities on **EKS (Kubernetes)** to parallelize data validation workloads.
- Enhanced observability using **CloudWatch Alarms**, **Lambda alerts**, and **custom Prometheus exporters**, improving SLA adherence and mean time to detect (MTTD) by 25%.
- Applied HIPAA-aligned data security with IAM roles, KMS encryption, Lake Formation, and column-level masking policies across all pipeline layers.
- Published gold-layer datasets to Tableau, improving dashboard performance by 30% and accelerating decision-making across clinical and claims analytics.

Tech Stack: Python, SQL, AWS Glue (PySpark), S3, Kinesis (Streams/Firehose), Redshift, Athena, Step Functions, EventBridge, Terraform, CloudFormation, Docker, Kubernetes (EKS), CloudWatch, Prometheus, IAM/KMS, Lake Formation, Tableau, Git/GitHub Actions

Excers.com – Data Engineer | Hyderabad, India | Mar 2020 – Oct 2021

Project: Retail Hadoop to AWS Redshift Migration (Data Lake & Warehouse)

- Migrated 1.5 TB/day of enterprise data from Hive to Amazon Redshift via Spark and Sqoop, **achieving 99.98% row-level parity (Hive → S3 → Redshift)**.
- Designed secure Parquet-based ingestion pipelines to **S3** with prefix-based partitioning and SSE-KMS encryption, meeting enterprise compliance requirements.
- Engineered incremental delta loading using staging tables and MERGE logic to enable reliable upserts and **reduce daily load times by 22%**.
- Tuned **Redshift COPY** and **VACUUM/ANALYZE** jobs using **DIST/SORT** keys and compression encodings, improving query latency by 35%.
- Automated data quality and reprocessing flows with file compaction, retries, and quarantine logic for malformed records, **cutting manual rework by 40%**.
- Containerized Spark ETL and testing scripts using **Docker**, enabling consistent local and CI/CD environments.
- Defined infrastructure using **Terraform and AWS CloudFormation**, automating provisioning of S3 buckets, Redshift clusters, and IAM roles across dev/test/prod.
- Deployed and scheduled ETL pipelines in **Airflow on Kubernetes**, leveraging Helm charts and ConfigMaps for parameterized job execution.
- Integrated CI/CD through **Jenkins pipelines** for ETL code testing, Docker image builds, and automated deployments to AWS.
- Established **CloudWatch and Prometheus-Grafana dashboards** for monitoring Redshift load performance and Airflow task SLAs, improving observability and incident response times by 25%.
- Conducted post-load QA using row-count checks, checksums, and aggregate comparisons to ensure data integrity across legacy and target systems.
- Supported UAT validation and KPI reconciliation across legacy Hive and modern AWS platforms, enabling smooth stakeholder sign-off.

Tech Stack: Python, SQL, Bash, Apache Spark, Hive, Sqoop, Amazon S3, Redshift, Parquet, Airflow (on Kubernetes), Terraform, CloudFormation, Docker, Jenkins, CloudWatch, Prometheus, Grafana, Git

EDUCATION:

- **University of North Texas, Denton, TX** — M.S. Data Science, **GPA 3.9** (Aug 2023 – May 2025)
- **KARE, Tamil Nadu, India** — B.S. Mechanical Engineering, **GPA 3.3** (Jun 2017 – Mar 2021)

CERTIFICATIONS:

- AWS Certified Data Engineer – Associate
- AWS Certified Machine Learning Engineer – Associate
- AWS Certified Cloud Practitioner
- Fundamentals of the Databricks Generative AI
- Prompt Engineering for ChatGPT by Vanderbilt University (Coursera)

TECHNICAL SKILLS:

Programming & Scripting	Python, SQL, Java, Scala, Bash/Shell, Linux
Data Engineering & ETL	Apache Airflow, AWS Glue, Azure Data Factory (ADF), dbt, Informatica, Talend, Apache NiFi, SSIS, Fivetran

Cloud Platforms & Storage	Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
Data Warehousing & Lakes	Snowflake, Amazon Redshift, Google BigQuery, Hadoop (HDFS)
Big Data & Stream Processing	Apache Spark (PySpark), Apache Hadoop, Azure Databricks, Apache Kafka, Apache Flume, Google Pub/Sub
Databases (SQL & NoSQL)	Oracle, MySQL, PostgreSQL, MongoDB, Cassandra, Spark SQL
BI & Visualization	Tableau, Power BI, SSRS
Monitoring & Observability	Grafana, CloudWatch, ELK Stack , Dynatrace, Quantum Metric
DevOps & CI/CD	Git, GitHub, Bitbucket, Jenkins, Docker, Kubernetes, Terraform, AWS CloudFormation
Data Governance & Quality	dbt Tests, AWS Deequ, Glue Data Quality, RBAC, Masking Policies
MLOps & Feature Engineering	Snowpark, SageMaker, MLflow, Feature Store Concepts, Model Monitoring
IDEs & Notebooks	Jupyter Notebook, IntelliJ IDEA, Visual Studio Code, Google Colab
Machine Learning & AI	Scikit-learn, spaCy, NLTK, Basic Deep Learning, LLM Concepts, Prompt Engineering